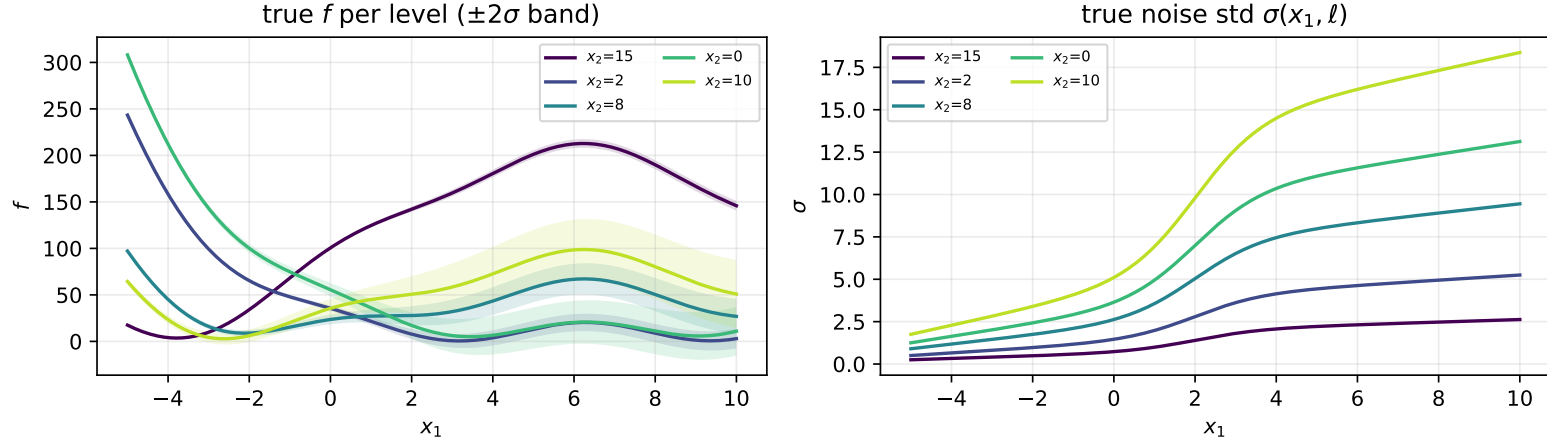


Test problems

Branin (heteroscedastic, 5 levels)

$$f(x_1, \ell) = \left(x_2(\ell) - \frac{5.1}{4\pi^2} x_1^2 + \frac{5}{\pi} x_1 - 6 \right)^2 + 10 \left(1 - \frac{1}{8\pi} \right) \cos x_1 + 10, \quad x_1 \in [-5, 10], \quad x_2(\ell) \in \{15, 2, 8, 0, 10\}$$

$$\sigma(x_1, \ell) = \left[0.5 + 0.15(x_1 + 5) + \frac{2.5}{1 + e^{-1.2(x_1 - 2)}} \right] m(\ell), \quad m(\ell) \in \{0.5, 1.0, 1.8, 2.5, 3.5\}$$

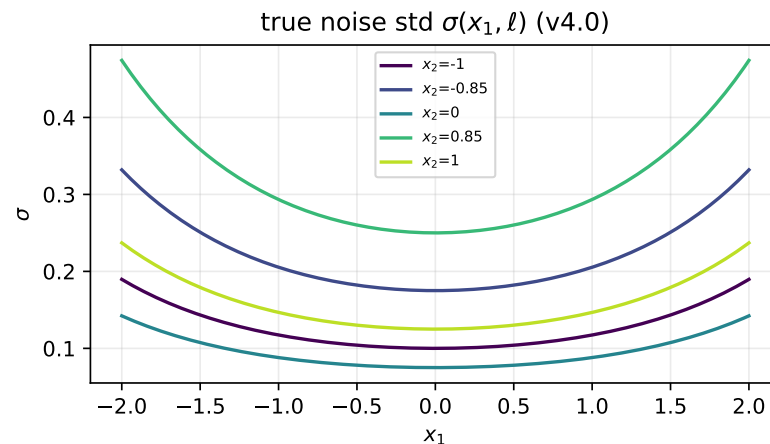
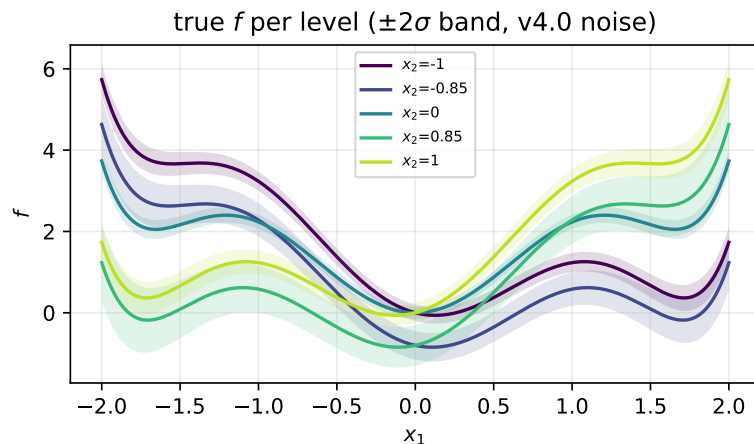


Six-hump camel (5 levels: two pairs + bridge)

$$f(x_1, \ell) = \left(4 - 2.1x_1^2 + \frac{x_1^4}{3} \right) x_1^2 + x_1 x_2(\ell) + (-4 + 4x_2(\ell)^2) x_2(\ell)^2, \quad x_1 \in [-2, 2], \quad x_2(\ell) \in \{-1.0, -0.85, 0.0, 0.85, 1.0\}$$

$$\sigma(x_1, \ell) = 0.05 e^{(0.4x_1)^2} m(\ell), \quad m(\ell) \in \{2.0, 3.5, 1.5, 5.0, 2.5\}$$

(noise as used in the 30-seed replication tables below)



Friedman 5-D + categorical (shape-only level pairs)

$$f_{\ell}(\mathbf{x}) = a(\ell) \cdot 10 \sin(\pi x_1 x_2) + 20(x_3 - 0.5)^2 + b(\ell) \cdot 10x_4 + 5x_5, \quad \mathbf{x} \in [0, 1]^5$$

$$a(\ell) \in \{1.0, 0.85, 0.5, 0.35\}, \quad b(\ell) \in \{1.0, 1.1, 1.5, 1.65\}, \quad \sigma(\mathbf{x}, \ell) = 0.1 \cdot 10^{x_1 - 0.5} m(\ell), \quad m(\ell) \in \{1.5, 1.0, 3.0, 2.0\}$$

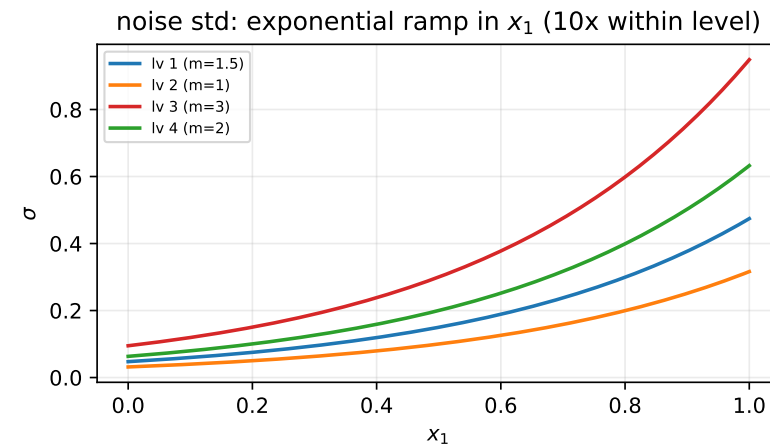
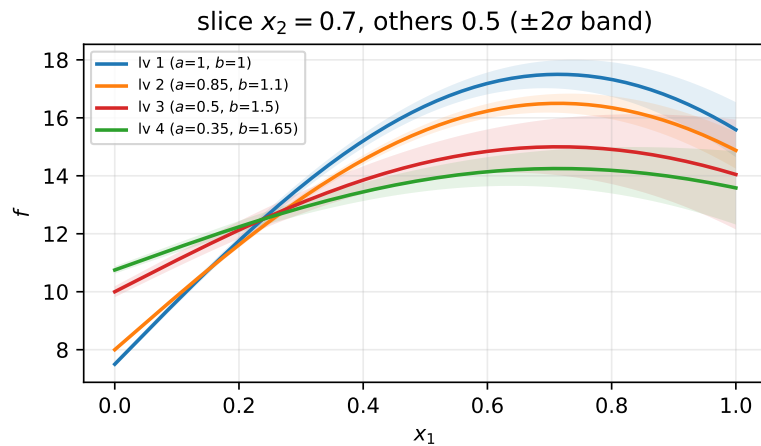


Table 1: 30-seed replication, all problems (fixed design, fresh noise replicates per seed): mean $\pm$ std over seeds. Bold on green = best mean per (problem, budget) block (coverage: closest to 0.95).

Problem	n	Model	MAE	RRMSE	IS	Coverage
Branin	25	Separate GPs	10.91 $\pm$ 0.466	0.434 $\pm$ 0.022	113.7 $\pm$ 11.60	0.775 $\pm$ 0.028
		Categorical GP	4.529 $\pm$ 0.241	0.159 $\pm$ 0.011	38.87 $\pm$ 2.018	0.946 $\pm$ 0.010
		Standard LVGP	4.419 $\pm$ 1.063	0.212 $\pm$ 0.051	74.80 $\pm$ 30.69	0.442 $\pm$ 0.143
		H-LVGP	2.520 $\pm$ 0.723	0.096 $\pm$ 0.037	37.28 $\pm$ 20.11	0.996 $\pm$ 0.016
		H-LVGP (torch)	3.752 $\pm$ 0.611	0.136 $\pm$ 0.029	24.61 $\pm$ 2.495	0.960 $\pm$ 0.023
	40	Separate GPs	4.454 $\pm$ 0.361	0.208 $\pm$ 0.017	42.88 $\pm$ 1.784	0.931 $\pm$ 0.011
		Categorical GP	1.585 $\pm$ 0.268	0.054 $\pm$ 0.012	14.86 $\pm$ 0.896	0.996 $\pm$ 0.010
		Standard LVGP	3.109 $\pm$ 0.705	0.122 $\pm$ 0.030	62.82 $\pm$ 20.09	0.410 $\pm$ 0.145
		H-LVGP	1.722 $\pm$ 0.425	0.064 $\pm$ 0.019	35.61 $\pm$ 10.83	0.998 $\pm$ 0.008
		H-LVGP (torch)	1.409 $\pm$ 0.195	0.048 $\pm$ 0.008	14.04 $\pm$ 1.000	0.976 $\pm$ 0.009
Six-hump camel	25	Separate GPs	0.231 $\pm$ 0.009	0.319 $\pm$ 0.007	2.072 $\pm$ 0.061	0.952 $\pm$ 0.008
		Categorical GP	0.168 $\pm$ 0.009	0.262 $\pm$ 0.010	1.778 $\pm$ 0.136	0.923 $\pm$ 0.012
		Standard LVGP	0.152 $\pm$ 0.018	0.242 $\pm$ 0.029	3.405 $\pm$ 0.521	0.417 $\pm$ 0.097
		H-LVGP	0.178 $\pm$ 0.012	0.287 $\pm$ 0.016	2.153 $\pm$ 0.231	0.905 $\pm$ 0.017
		H-LVGP (torch)	0.231 $\pm$ 0.019	0.344 $\pm$ 0.022	2.211 $\pm$ 0.224	0.917 $\pm$ 0.022
	40	Separate GPs	0.196 $\pm$ 0.008	0.295 $\pm$ 0.006	2.313 $\pm$ 0.111	0.888 $\pm$ 0.015
		Categorical GP	0.114 $\pm$ 0.010	0.160 $\pm$ 0.015	1.127 $\pm$ 0.164	0.918 $\pm$ 0.013
		Standard LVGP	0.145 $\pm$ 0.034	0.212 $\pm$ 0.066	3.240 $\pm$ 1.092	0.324 $\pm$ 0.106
		H-LVGP	0.143 $\pm$ 0.024	0.226 $\pm$ 0.042	1.147 $\pm$ 0.318	0.953 $\pm$ 0.029
		H-LVGP (torch)	0.174 $\pm$ 0.017	0.237 $\pm$ 0.019	1.449 $\pm$ 0.214	0.915 $\pm$ 0.022
Friedman 5-D	40	Separate GPs	2.656 $\pm$ 0.015	0.664 $\pm$ 0.004	23.97 $\pm$ 0.275	0.716 $\pm$ 0.009
		Categorical GP	0.883 $\pm$ 0.011	0.239 $\pm$ 0.003	7.819 $\pm$ 0.256	0.873 $\pm$ 0.007
		Standard LVGP	1.218 $\pm$ 0.227	0.327 $\pm$ 0.051	15.52 $\pm$ 4.699	0.683 $\pm$ 0.102
		H-LVGP	1.242 $\pm$ 0.183	0.332 $\pm$ 0.042	14.92 $\pm$ 3.072	0.712 $\pm$ 0.067
		H-LVGP (torch)	1.239 $\pm$ 0.175	0.329 $\pm$ 0.039	14.19 $\pm$ 2.913	0.727 $\pm$ 0.063
	60	Separate GPs	1.858 $\pm$ 0.047	0.480 $\pm$ 0.011	16.44 $\pm$ 1.483	0.768 $\pm$ 0.022
		Categorical GP	0.483 $\pm$ 0.028	0.135 $\pm$ 0.008	3.286 $\pm$ 0.223	0.963 $\pm$ 0.007
		Standard LVGP	0.505 $\pm$ 0.083	0.138 $\pm$ 0.022	4.128 $\pm$ 0.637	0.799 $\pm$ 0.048
		H-LVGP	0.552 $\pm$ 0.131	0.154 $\pm$ 0.035	3.529 $\pm$ 1.010	0.910 $\pm$ 0.019
		H-LVGP (torch)	0.504 $\pm$ 0.028	0.143 $\pm$ 0.010	3.473 $\pm$ 0.358	0.897 $\pm$ 0.018
	80	Separate GPs	1.274 $\pm$ 0.014	0.345 $\pm$ 0.004	8.091 $\pm$ 0.122	0.919 $\pm$ 0.004
		Categorical GP	0.317 $\pm$ 0.009	0.101 $\pm$ 0.003	2.150 $\pm$ 0.095	0.962 $\pm$ 0.005
		Standard LVGP	0.348 $\pm$ 0.048	0.100 $\pm$ 0.014	2.638 $\pm$ 0.594	0.820 $\pm$ 0.043
		H-LVGP	0.345 $\pm$ 0.021	0.101 $\pm$ 0.006	2.018 $\pm$ 0.123	0.973 $\pm$ 0.013
		H-LVGP (torch)	0.364 $\pm$ 0.014	0.107 $\pm$ 0.003	2.345 $\pm$ 0.173	0.900 $\pm$ 0.017

Table 2: 30-seed replication, all problems (fixed design, fresh noise replicates per seed): mean $\pm$ std over seeds. Bold on green = best mean per (problem, budget) block (coverage: closest to 0.95). Without the Categorical GP (winners recomputed).

Problem	n	Model	MAE	RRMSE	IS	Coverage
Branin	25	Separate GPs	10.91 ± 0.466	0.434 ± 0.022	113.7 ± 11.60	0.775 ± 0.028
		Standard LVGP	4.419 ± 1.063	0.212 ± 0.051	74.80 ± 30.69	0.442 ± 0.143
		H-LVGP	2.520 ± 0.723	0.096 ± 0.037	37.28 ± 20.11	0.996 ± 0.016
		H-LVGP (torch)	3.752 ± 0.611	0.136 ± 0.029	24.61 ± 2.495	0.960 ± 0.023
	40	Separate GPs	4.454 ± 0.361	0.208 ± 0.017	42.88 ± 1.784	0.931 ± 0.011
		Standard LVGP	3.109 ± 0.705	0.122 ± 0.030	62.82 ± 20.09	0.410 ± 0.145
		H-LVGP	1.722 ± 0.425	0.064 ± 0.019	35.61 ± 10.83	0.998 ± 0.008
		H-LVGP (torch)	1.409 ± 0.195	0.048 ± 0.008	14.04 ± 1.000	0.976 ± 0.009
Six-hump camel	25	Separate GPs	0.231 ± 0.009	0.319 ± 0.007	2.072 ± 0.061	0.952 ± 0.008
		Standard LVGP	0.152 ± 0.018	0.242 ± 0.029	3.405 ± 0.521	0.417 ± 0.097
		H-LVGP	0.178 ± 0.012	0.287 ± 0.016	2.153 ± 0.231	0.905 ± 0.017
		H-LVGP (torch)	0.231 ± 0.019	0.344 ± 0.022	2.211 ± 0.224	0.917 ± 0.022
	40	Separate GPs	0.196 ± 0.008	0.295 ± 0.006	2.313 ± 0.111	0.888 ± 0.015
		Standard LVGP	0.145 ± 0.034	0.212 ± 0.066	3.240 ± 1.092	0.324 ± 0.106
		H-LVGP	0.143 ± 0.024	0.226 ± 0.042	1.147 ± 0.318	0.953 ± 0.029
		H-LVGP (torch)	0.174 ± 0.017	0.237 ± 0.019	1.449 ± 0.214	0.915 ± 0.022
Friedman 5-D	40	Separate GPs	2.656 ± 0.015	0.664 ± 0.004	23.97 ± 0.275	0.716 ± 0.009
		Standard LVGP	1.218 ± 0.227	0.327 ± 0.051	15.52 ± 4.699	0.683 ± 0.102
		H-LVGP	1.242 ± 0.183	0.332 ± 0.042	14.92 ± 3.072	0.712 ± 0.067
		H-LVGP (torch)	1.239 ± 0.175	0.329 ± 0.039	14.19 ± 2.913	0.727 ± 0.063
	60	Separate GPs	1.858 ± 0.047	0.480 ± 0.011	16.44 ± 1.483	0.768 ± 0.022
		Standard LVGP	0.505 ± 0.083	0.138 ± 0.022	4.128 ± 0.637	0.799 ± 0.048
		H-LVGP	0.552 ± 0.131	0.154 ± 0.035	3.529 ± 1.010	0.910 ± 0.019
		H-LVGP (torch)	0.504 ± 0.028	0.143 ± 0.010	3.473 ± 0.358	0.897 ± 0.018
	80	Separate GPs	1.274 ± 0.014	0.345 ± 0.004	8.091 ± 0.122	0.919 ± 0.004
		Standard LVGP	0.348 ± 0.048	0.100 ± 0.014	2.638 ± 0.594	0.820 ± 0.043
		H-LVGP	0.345 ± 0.021	0.101 ± 0.006	2.018 ± 0.123	0.973 ± 0.013
		H-LVGP (torch)	0.364 ± 0.014	0.107 ± 0.003	2.345 ± 0.173	0.900 ± 0.017